



NORTH SYDNEY GIRLS HIGH SCHOOL

HSC MATHEMATICS ASSESSMENT TASK

TERM 1 – 2006

Time Allowed: 1 hour + 2 minutes reading time

Instructions:

- Start each question on a new page
- Write on one side of the paper only, work down the page and do not work in columns
- Leave a margin on the left hand side of the page
- Show all necessary working
- Marks may not be awarded for untidy or poorly arranged work
- Diagrams are not drawn to scale
- There are five questions
- Marks are as indicated

This task is worth 20% of the HSC Assessment Mark

Name: _____

Question 1 (10 marks)	Marks
(a) Find the third term of the sequence whose n^{th} term is given by $T_n = 3 \times 2^{n-2}$	1
(b) Consider the following series $101 + 96 + 91 + \dots$	
(i) Explain why this series is arithmetic.	1
(ii) State the next term.	1
(iii) Which term is equal to 26?	2
(iv) Find the sum of the first 10 terms.	2
(c) Evaluate $\sum_{r=1}^4 \frac{1}{r}$	1
(d) A geometric series has T_5 equal to $\frac{81}{8}$ and T_2 equal to 3. Find an expression for T_n .	2

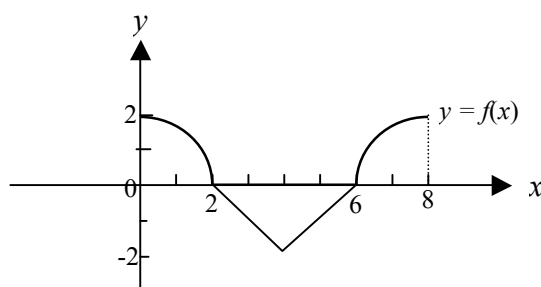
Question 2 (11 marks) Start a new page.	Marks
(a) A given geometric series has a limiting sum of 36 and its first term is 27. Find the common ratio.	3
(b) A retired woman decides to live off her savings. She has \$70 000 and invests it at an interest rate of 6% per annum, compounded monthly. At the end of each month after interest has been received, she withdraws \$ D . Let the amount of money left at the end of the n^{th} month just after she has made her withdrawal be \$ A_n	
(i) Find an expression for A_1 and use it to show that $A_2 = 70\,000(1.005)^2 - D(1.005 + 1)$	2
(ii) Write down an expression for A_n , the amount of money left after n months.	2
(iii) Show that $D = \frac{70\,000(1.005)^n - A_n}{\left[\frac{(1.005)^n - 1}{0.005} \right]}$	2
(iv) Find the monthly withdrawal, D , if the woman has no money left after 10 years.	2

Question 3 (9 marks) **Start a new page.****Marks**

a) Find the indefinite integral of $(5x+1)^2$ **3**

b) Evaluate $\int_{25}^{36} \frac{1}{\sqrt{x}} dx$ **3**

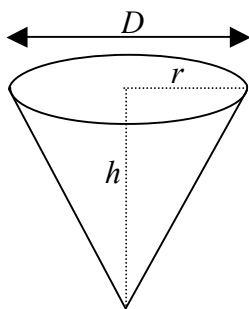
c) The graph of $y = f(x)$ is shown below. It consists of two circular arcs and intervals.



Evaluate $\int_0^8 f(x) dx$ **3**

Question 4 (8 marks) **Start a new page.**

The diagram below represents a conical water container.



In this cone, the sum of the base diameter, D , and the height, h , is 60 metres.

a) Write an expression for the height, h , in terms of the radius, r . **2**

b) Show that the volume of the container is given by

$$V = 20\pi r^2 - \frac{2}{3}\pi r^3$$

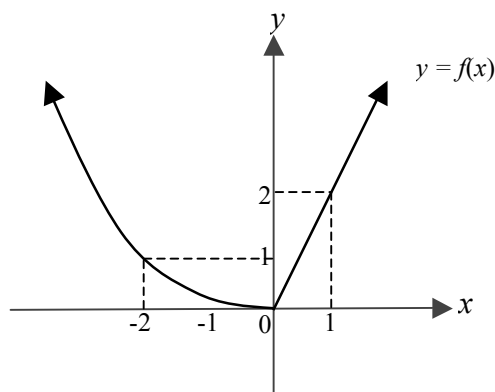
[You may use the formula $V = \frac{1}{3}\pi r^2 h$, the volume of a cone] **2**

c) Find the radius that makes the volume of the container a maximum. **4**

Question 5 (11 marks) **Start a new page.**

Marks

- a) The graph of $y = f(x)$ is drawn below



On your own paper, sketch the graph of $y = f'(x)$.

3

- b) Consider the curve $f(x) = 7 + 4x^3 - 3x^4$

- i) Given that $y = f(x)$ has a stationary point of inflexion at $(0, 7)$, find any other stationary point(s) and determine their(its) nature.

3

- ii) The graph of $y = f(x)$ passes through the point $(\frac{2}{3}, 7\frac{16}{27})$. Show that this point is a point of inflexion.

2

- iii) Sketch the graph of $y = f(x)$ showing stationary points, points of inflexion and the intercept on the y-axis.

3

End of paper

Solutions

QUESTION 1

1a) $T_3 = 6$

b) (i) $T_3 - T_2 = -5$
 $T_2 - T_1 = -5$
 So the
 common difference
 d is equal

(ii) 86

(iii) $T_n = a + (n-1)d$
 $T_n = 101 + (n-1) \times -5$
 $= 106 - 5n$

Let $106 - 5n = 26$
 $5n = 80$
 $n = 16$

T_{16}

(iv) $S_{10} = \frac{10}{2} \{202 + 9 \times -5\}$
 $= 5 \{202 - 45\}$
 $= 785$

c) $2\frac{1}{12}$

d) $ar^4 = \frac{81}{8}$ (1)
 $ar = 3$ (2)

(1) $\div$ (2)
 $r^3 = \frac{27}{8}$

$r = \frac{3}{2}$

FROM (2),

$a \times \frac{3}{2} = 3$

$a = 2$

$T_n = 2\left(\frac{3}{2}\right)^{n-1}$

OR $T_n = 2^{2-n} \cdot 3^{n-1}$

QUESTION 2

a) $36 = \frac{27}{1-r}$

$36(1-r) = 27$

$1-r = \frac{3}{4}$

$r = \frac{1}{4}$

b)

(i) $A_1 = 70000(1.005) - D$

$A_2 = \{70000(1.005) - D\} \cdot 1.005 - D$
 $= 70000(1.005)^2 - 1.005D - D$
 $= 70000(1.005)^2 - D(1.005 + 1)$

(ii)

$A_n = 70000(1.005)^n - D(1 + 1.005 + \dots + (1.005)^{n-1})$

(iii)

$A_n = 70000(1.005)^n - D \left[\frac{(1.005)^n - 1}{0.005} \right]$

$D \left[\frac{(1.005)^n - 1}{0.005} \right] = 70000(1.005)^n - A_n$

$D = \frac{70000(1.005)^n - A_n}{\left[\frac{(1.005)^n - 1}{0.005} \right]}$

(iv) Let $A_n = 0$ and $n = 120$

$D = 70000(1.005)^{120} \times \frac{0.005}{(1.005)^{120} - 1}$

$= \$777.14(3)$

QUESTION 3

$$a) \int (5x+1)^2 dx = \frac{(5x+1)^3}{15} + C \quad \text{or} \quad \int (25x^2 + 10x + 1) dx = \frac{25x^3}{3} + 5x^2 + x + C$$

$$\begin{aligned} b) \int_{25}^{36} \frac{1}{\sqrt{x}} dx &= \int_{25}^{36} x^{-\frac{1}{2}} dx \\ &= 2 \left[x^{\frac{1}{2}} \right]_{25}^{36} \\ &= 2 \left[\sqrt{x} \right]_{25}^{36} \\ &= 2 \left[\sqrt{36} - \sqrt{25} \right] \\ &= 2 \left[6 - 5 \right] \\ &= 2 \end{aligned}$$

$$\begin{aligned} c) \int_0^9 f(x) dx &= \text{AREA OF 2 QUADRANTS} + \text{'NEGATIVE' AREA OF TRIANGLE} \\ &= \frac{1}{2} \pi r^2 + \frac{1}{2} \times 4 \times 2 \\ &= 2\pi - 4 \end{aligned}$$

QUESTION 4

$$2r + h = 60$$

$$a) h = 60 - 2r$$

$$\begin{aligned} b) V &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \pi r^2 (60 - 2r) \\ &= 20\pi r^2 - \frac{2}{3} \pi r^3 \end{aligned}$$

$$c) \frac{dV}{dr} = 40\pi r - 2\pi r^2$$

$$\begin{aligned} \frac{dV}{dr} &= 0 \quad \text{Put } \frac{dV}{dr} = 0 \\ 40\pi r - 2\pi r^2 &= 0 \\ 2\pi r(20 - r) &= 0 \\ r &= 20 \end{aligned}$$

Now

$$\frac{d^2V}{dr^2} = 40\pi - 4\pi r$$

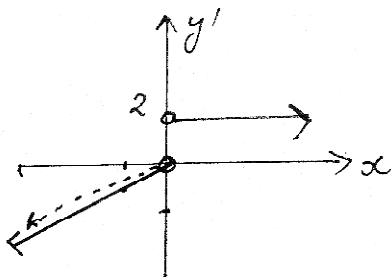
$$\text{at } r = 20$$

$$\frac{d^2V}{dr^2} = 40\pi - 80\pi = -40\pi$$

$< 0 \Rightarrow \text{MAX VALUE}$

QUESTION 5

a)



$$b) f(x) = 7 + 4x^3 - 3x^4$$

$$f'(x) = 12x^2 - 12x^3$$

$$= 12x^2(1-x)$$

$$\text{Put } f'(x) = 0$$

$$12x^2(1-x) = 0$$

$$x = 0, 1$$

Consider $x = 1$

$$(1, 8)$$

$$f''(x) = 24x - 36x^2$$

$$f''(1) = -12 < 0 \Rightarrow$$

CONCAVE DOWN
i.e. MAXIMUM
TURNING
POINT

$(1, 8)$ is MAXIMUM TURNING
POINT

(II) POINT OF INFLECTION

$$f''(x) = 24x - 36x^2$$

$$\text{Put } f''(x) = 0$$

$$24x - 36x^2 = 0$$

$$12x(2-3x) = 0$$

$$x = 0, \frac{2}{3}$$

Check $x = \frac{2}{3}$

$$f''\left(\frac{1}{2}\right) = 3 > 0 \Rightarrow \text{concave upward}$$

$$f''\left(\frac{2}{3}\right) = -2\frac{1}{2} < 0 \Rightarrow \text{CONCAVE DOWNWARD}$$

$$\text{OR } f''(1) = -12 < 0$$

$\left(\frac{2}{3}, 7\frac{16}{27}\right)$ is a
POINT OF
INFLECTION

(III)

